

Fisher Scoring and MCMC for GLMs

Probability and Statistics – Final Project (Group 19)

1 Generalized Linear Models and Algorithms

1.1 Generalized Linear Models (GLM)

A GLM extends classical linear regression to allow response variables from the exponential family. It comprises three components:

1. Random component: $Y_i \sim f(y_i; \theta_i, \phi) = \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\right\}$
2. Systematic component: $\eta_i = X_i^\top \beta$
3. Link function: $g(\mu_i) = \eta_i$ where $\mu_i = \mathbb{E}[Y_i | X_i]$

For canonical link functions, $\theta_i = \eta_i = X_i^\top \beta$.

1.2 Fisher Scoring Algorithm

1.2.1 Algorithm Description

Fisher Scoring iteratively updates parameter estimates as follows:

1. Initialize the parameter vector β_0 (typically as a zero vector).
2. For iteration $k = 0, 1, 2, \dots$ until convergence, update $\beta_{k+1} = \beta_k + [I(\beta_k)]^{-1} \nabla \ell(\beta_k)$, where $\nabla \ell(\beta_k)$ denotes the score function and $I(\beta_k)$ is the Fisher information matrix.
3. Check convergence using a criterion such as $\|\nabla \ell(\beta_k)\| < 10^{-8}$.

1.2.2 Gradient of the Log-Likelihood

The log-likelihood is $\ell(\beta) = \sum_{i=1}^n \left[\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right]$.

For exponential families, $\mu_i = b'(\theta_i)$ $\text{Var}(Y_i | x_i) = b''(\theta_i)a(\phi)$

$$\frac{\partial \ell(\beta)}{\partial \beta_j} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \sum_{i=1}^n \frac{y_i - \mu_i}{a(\phi)} \frac{a(\phi)}{\text{Var}(Y_i | x_i)} \frac{1}{g'(\mu_i)} x_{ij} = \sum_{i=1}^n \frac{(y_i - \mu_i)x_{ij}}{\text{Var}(Y_i | x_i) g'(\mu_i)}.$$

In matrix form, $\nabla \ell(\beta) = X^\top A(y - \mu)$, where $A = \text{diag}\left(\frac{1}{\text{Var}(Y_i | x_i) g'(\mu_i)}\right)$.

For canonical link function, $\theta_i = \eta_i$, so $\frac{\partial \ell(\beta)}{\partial \beta_j} = \sum_{i=1}^n \frac{y_i - \mu_i}{a(\phi)} x_{ij}$ or $\nabla \ell(\beta) = \frac{1}{a(\phi)} X^\top (y - \mu)$.

For Bernoulli and Poisson models in particular, $a(\phi) = 1$, and hence $\nabla \ell(\beta) = X^\top (y - \mu)$.

1.2.3 Fisher Information Matrix

The (j, k) -th element of the Fisher information matrix is defined as

$$I_{jk}(\beta) = \mathbb{E}\left[\frac{\partial \ell(\beta)}{\partial \beta_j} \frac{\partial \ell(\beta)}{\partial \beta_k}\right] = \mathbb{E}\left[\left(\sum_{i=1}^n \frac{(y_i - \mu_i)x_{ij}}{\text{Var}(Y_i) g'(\mu_i)}\right) \left(\sum_{i=1}^n \frac{(y_i - \mu_i)x_{ik}}{\text{Var}(Y_i) g'(\mu_i)}\right)\right] = \sum_{i=1}^n \frac{x_{ij}x_{ik}}{\text{Var}(Y_i) [g'(\mu_i)]^2}$$

In matrix form, $I(\beta) = X^\top W X$, where $W = \text{diag}(w_1, \dots, w_n)$, $w_i = \frac{1}{\text{Var}(Y_i) [g'(\mu_i)]^2}$

Logistic regression (logit link): $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$ and $g'(\mu_i) = \frac{1}{\mu_i(1 - \mu_i)}$, so $w_{ii} = \mu_i(1 - \mu_i)$.

Poisson regression (log link): $\text{Var}(Y_i) = \mu_i$ and $g'(\mu_i) = \frac{1}{\mu_i}$, so $w_{ii} = \mu_i$.

1.2.4 Equivalence of Fisher Scoring and Newton's Method under Canonical Links

Newton's method updates $\beta^{(k+1)} = \beta^{(k)} - (\nabla^2 \ell(\beta^{(k)}))^{-1} \nabla \ell(\beta^{(k)})$, while Fisher Scoring replaces $-\nabla^2 \ell$ by $I(\beta) = \mathbb{E}[-\nabla^2 \ell(\beta)]$.

Under a canonical link, $\theta_i = x_i^T \beta$, so $\mu_i = b'(x_i^T \beta)$ and $\nabla^2 \ell(\beta) = -\sum_i \frac{b''(x_i^T \beta)}{a(\phi)} x_i x_i^T$, which depends only on β and X , not on the random data y_i .

Hence $\mathbb{E}[\nabla^2 \ell(\beta)] = \nabla^2 \ell(\beta)$, proving the two methods coincide.

1.3 MCMC - Random Walk Metropolis-Hastings (RWMH)

1.3.1 Algorithm Description

Under the Bayesian framework, we use $p(\beta|y, X) \propto p(y|X, \beta) p(\beta)$ for inference. When direct sampling is infeasible, the RWMH algorithm provides a Markov chain Monte Carlo (MCMC) approach, i.e. constructs a Markov chain with stationary distribution equal to the posterior:

1. Choose an initial $\beta^{(0)}$ and a symmetric proposal distribution, e.g. $q(\beta^* | \beta^{(t)}) = \mathcal{N}(\beta^{(t)}, \Sigma)$.
2. At iteration $t + 1$, draw candidate $\beta^* \sim q(\beta^* | \beta^{(t)})$.
3. Compute the acceptance probability $\alpha = \min\left(1, \frac{p(\beta^* | y, X)}{p(\beta^{(t)} | y, X)}\right)$.
4. Draw $u \sim \mathcal{U}(0, 1)$. If $u \leq \alpha$, set $\beta^{(t+1)} = \beta^*$; otherwise $\beta^{(t+1)} = \beta^{(t)}$.

1.3.2 Acceptance Probability Derivation

In RWMH, the proposal distribution is symmetric and satisfies $q(\beta^* | \beta^{(t)}) = q(\beta^{(t)} | \beta^*)$, so the acceptance probability simplifies to $\alpha = \min\{1, \frac{p(\beta^* | y, X)}{p(\beta^{(t)} | y, X)}\}$.

For numerical stability, the acceptance probability is evaluated on the logarithmic scale. A proposal is accepted if $\log u < \log p(\beta^* | y, X) - \log p(\beta^{(t)} | y, X)$.

1.3.3 Discussion of the Prior Distributions Chosen

For both datasets, we used independent Gaussian priors $\beta \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$.

1. Weakly Informative Nature. Setting large variances (20.0 for **CreditCard**, 25.0 for **azcabgptca**) so the prior is weakly informative and the posterior is driven mainly by the data.
2. Symmetry and Neutrality. Centering at zero encodes no directional preference a priori, letting the sign and magnitude of covariate effects be determined by the observed data.

1.4 Key Differences Induced by Non-Canonical Link Functions

1.4.1 Loss of Equivalence Between Fisher Scoring and Newton's Method

$$\frac{\partial^2 \ell(\beta)}{\partial \beta_j \partial \beta_k} = \sum_{i=1}^n \left[\frac{\partial}{\partial \beta_k} (y_i - \mu_i) \right] \frac{x_{ij}}{\text{Var}(Y_i) g'(\mu_i)} + \sum_{i=1}^n (y_i - \mu_i) \frac{\partial}{\partial \beta_k} \left(\frac{1}{\text{Var}(Y_i) g'(\mu_i)} \right) x_{ij}$$

Under a canonical link function, $\frac{1}{\text{Var}(Y_i) g'(\mu_i)} = \frac{1}{a(\phi)}$ (a constant), hence the second term vanishes. The Hessian does not depend on y_i so $\mathbb{E}[H(\beta)] = H(\beta)$ and Fisher scoring coincides with Newton's method. Under a non-canonical one, this term is no longer constant. The Hessian depends on y_i , so $\mathbb{E}[H(\beta)] \neq H(\beta)$ and Fisher scoring and Newton's method no longer coincide.

1.4.2 Increased Complexity of the Weight Matrix

Under a canonical link, the diagonal weight matrix $w_i = \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 = \frac{b''(\theta_i)}{a(\phi)} = \frac{\text{Var}(Y_i)}{a(\phi)^2}$.

Under a non-canonical link, the weight matrix becomes more complex. For instance, in the probit regression model, $\mu_i = \Phi(\eta_i)$, $\text{Var}(Y_i) = \mu_i(1 - \mu_i) = \Phi(\eta_i)(1 - \Phi(\eta_i))$, $\frac{\partial \mu_i}{\partial \eta_i} = \phi(\eta_i)$.

The corresponding weight is then $w_i = \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 = \frac{\phi(\eta_i)^2}{\Phi(\eta_i)(1 - \Phi(\eta_i))}$.

2 Credit Card Dataset Analysis

Variable Exclusion: We excluded `expenditure` and `share` as they represent post-approval behavior, introducing data leakage and potential reverse causality.

Fisher Scoring: The Fisher scoring algorithm converged in 7 iterations. Convergence plots of the parameter estimates and log-likelihood values indicate stable and monotonic convergence.

MCMC (Random Walk Metropolis–Hastings): The trace plot for income shows stable fluctuations with no trend. The ACF decays rapidly (within 7 lags), indicating good mixing and adequate effective sample size.

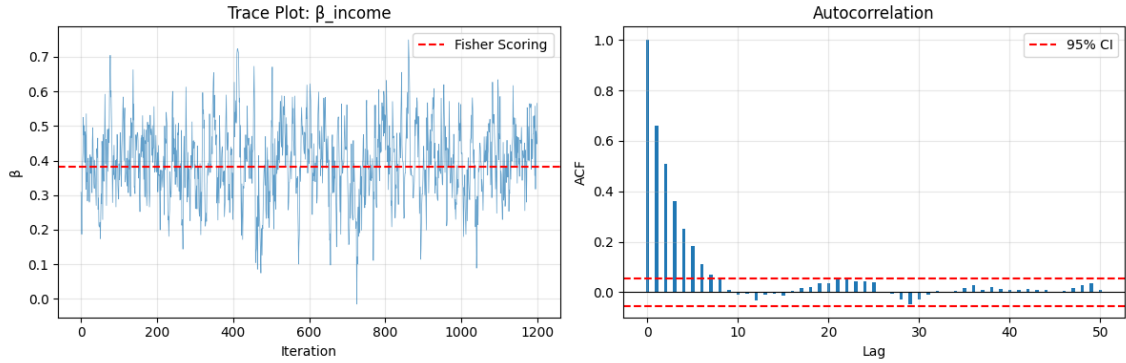


Figure 1: MCMC diagnostics for the income coefficient.

Final Discussion on the Fitted Models

Both approaches produce consistent results with similar coefficients and confidence and credible intervals overlap, confirming methodological robustness.

Method	Intercept	reports	age	income	owner	selfemp	dependents	months	majorcards	active
Fisher (Coef)	1.2998	-2.3565	-0.1269	0.3833	0.4783	-0.7573	-0.3023	0.0338	0.1954	0.8342
MCMC (Mean)	1.2889	-2.3552	-0.1224	0.3958	0.5138	-0.7061	-0.3101	0.0302	0.1921	0.8252
MCMC (Std)	0.1300	0.1689	0.0959	0.1105	0.2108	0.2530	0.0840	0.0887	0.0767	0.1114

In the model, coefficients describe the effect of covariates on the log-odds of credit card acceptance: positive coefficients increase the odds of approval, *ceteris paribus* and vice versa.

In our analysis, we initially used standardized covariates to identify the most influential factors by comparing the magnitudes of their coefficients. To provide a practical interpretation for the final discussion, we back-transformed these coefficients (β_{std}) to their **original units** using the formula $\beta_{raw} = \frac{\beta_{std}}{\sigma_x}$, where σ_x is the sample standard deviation of the original variable. This back-transformation is applied only to non-binary covariates, while binary indicators are interpreted directly. Effects are then reported in terms of the **Odds Ratio (OR)** for logistic regression or the **Incidence Rate Ratio (IRR)** for Poisson regression, computed as $\exp(\beta_{raw})$.

Overall, the results suggest that credit card approval is primarily driven by indicators of creditworthiness and financial reliability rather than demographic characteristics. The three most influential covariates are:

1. **reports** (-2.3565): Each additional derogatory report reduces approval odds by 82.65%.
2. **active** (0.8342): Each additional active credit account increases approval odds by 14.14%.
3. **selfemp** (-0.7573): Being self-employed reduces approval odds by 53.11% relative to employed applicants.

3 azcabgptca Dataset Analysis

Fisher Scoring: The Fisher scoring algorithm converged in 15 iterations. Convergence diagnostics based on the log-likelihood and gradient norm show stable convergence despite mild non-monotonic behavior in the early iterations.

MCMC (Random Walk Metropolis–Hastings): The trace plot for procedure shows stable fluctuations with no trend. The ACF decays rapidly (within 10 lags), and despite minor ripples around lag 30, the overall low values indicate good convergence and sample independence.

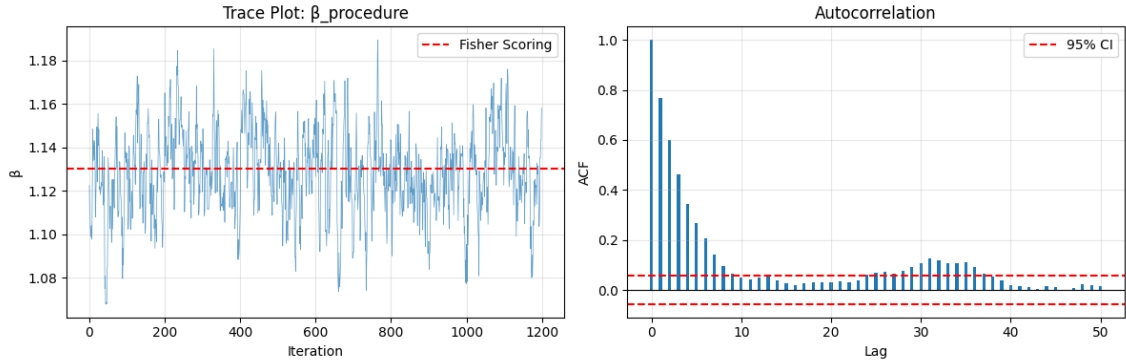


Figure 2: MCMC diagnostics for the procedure coefficient.

Final Discussion on the Fitted Models

Both approaches lead to consistent results regarding the determinants of hospital length of stay.

Method	Intercept	died	procedure	age	gender	type
Fisher (Coef)	1.2821	-0.0413	1.1301	0.0432	-0.1061	0.1894
MCMC (Mean)	1.2854	-0.0413	1.1282	0.0433	-0.1084	0.1891
MCMC (Std)	0.0216	0.0081	0.0196	0.0084	0.0182	0.0178

In the model, coefficients describe the effect of covariates on the log of the expected hospital length of stay: positive coefficients increase the expected duration, *ceteris paribus* and vice versa. Overall, the results suggest that clinical factors, especially procedural invasiveness, play a dominant role in explaining hospital stay duration. The three most influential covariates are:

1. **procedure** (1.1301): Undergoing a CABG procedure is associated with an 209.61% increase in the expected duration of hospitalization relative to PTCA
2. **type** (0.1894): Compared to elective cases (*type*=0), emergency or urgent admissions (*type*=1) increases the expected hospital length of stay by 20.85%.
3. **gender** (-0.1061): Male patients have an expected hospital stay that is approximately 10.06% shorter than female ones do.